

LIVE-GS: LLM Powers Interactive VR by Enhancing Gaussian Splatting

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ABSTRACT

We propose LIVE-GS, a highly realistic interactive Gaussian splatting system in VR environments powered by LLM. Our pipeline supports reconstructions and physically-based interactions in VR, integrating object-aware reconstruction, GPT-assisted inpainting, and a computationally efficient simulation framework. To enhance scene understanding, we prompt GPT-4o to analyze the physical properties of objects in the scene, thereby guiding physical simulations to align with real-world phenomena. Our experimental results demonstrate that with the assistance of LLM's understanding and scene enhancement, our VR system can support complex and realistic interactions without requiring additional manual design or annotation.

Index Terms: 3D Gaussian Splatting, virtual reality, material analysis

1 INTRODUCTION

3D Gaussian Splatting (3DGS) [2] has quickly taken precedence for its remarkable novel synthesis, real-time efficiency, and seamless integration with the modern rasterization pipeline. VR-GS [1] successfully implements the first interactive physics-based VR system based on 3DGS, demonstrating its potential for interactive VR scene reconstruction. However, their method limits interactions to elastic deformation due to the absence of semantics, leading to potentially unrealistic experiences. Although approaches like Physdreamer [5] attempt to reconstruct physical properties, they are constrained by pre-assumptions and struggle with complex real-world scenes.

To address these limitations, we introduce LIVE-GS, an interactive VR system based on 3DGS, capable of performing physical property analysis tasks. Leveraging GPT models, our system demonstrates strong scene understanding, enabling the analysis of physical properties from minimal input images. Our system aims to reconstruct complex real-world scenes and facilitate real-time interactions consistent with actual phenomena. Experiments show that our system delivers competitive results in simulation quality, low latency, and effective analysis for VR interactions.

2 METHOD

We introduce our pipeline in three parts. Through our pipeline, our system constructs the scene with 3DGS, labels each object, segments them, and assigns physical properties based on analysis results, enabling highly efficient interactions in VR.

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2.1 Instance-aware Scene Reconstruction

Gaussian training with identity encoding 3DGS [2] represents a scene using a collection of colored 3D Gaussians and renders them into camera views through splatting-based rasterization. We introduce additional identity features to each kernel and jointly train them for segmentation. For enhanced robustness and accuracy, a grouping loss that includes **2D identity loss** and **3D regularization loss** is introduced from Gaussian Grouping [4].

Feature-mask segmentation To segment objects from environments for subsequent processing, we design a feature-mask method including a *feature stage* and a *mask stage*. In the feature stage, we choose a classifier with 3D regularization loss. In the mask stage, we project kernels to screen space and decide the segmented parts according to the mask images in the other stage. We finally take the intersection of results in two stages as an outcome.

2.2 GPT-powered Scene Enhancement

We design strategies to leverage GPT to enhance scene understanding of our system, which consists of **material analysis** and **artifact tracking**.

Material Analysis In our system, we use GPT-4o for material analysis due to its strong visual perception capabilities. The model is required to accept images and distinguish between deformation, granular material, and rigid, estimating their properties. These can be summarized into two categories: **object property** and **particle property**.

Object property This kind of property is volume-dependent, which can be affected when analyzed under perspective transformations. Therefore, instead of applying them directly to our engine, we supplement a correction factor C related to the particle count to resolve the above problems.

Particle property This kind of property pertains to constraints between particles, which is independent of the object size. These values are required to be normalized, which favours the judgment of our system. These attributes are then converted proportionally in certain ranges and input into our system:

1. Deformation: *Deformation Resistance* and *Plasticity*.
2. Granular material: *Friction*.
3. Rigid: *Fragility* and *Force Threshold*.

Artifact tracking Unseen regions may be exposed after object removal and generate artifacts. We use GPT-4o to analyze the color of the artifact regions and use DEVA to track them. Subsequently, we insert extra Gaussian kernels in the removed region and freeze other kernels during a fine-training stage with the 2D inpainting method. Considering the irregularity of the mask, the SSIM loss is discarded here for better results.

2.3 Interactive Physical Simulation Framework

For real-time rendering and interaction in the VR system, we implement our simulation with an interpolation method based on Obi solver [3]. We generate physical particles on object surfaces and bind Gaussian kernels to the closest ones. Kernels' changes in rotation and translation can be calculated through interpolation weighted by distances to related particles.

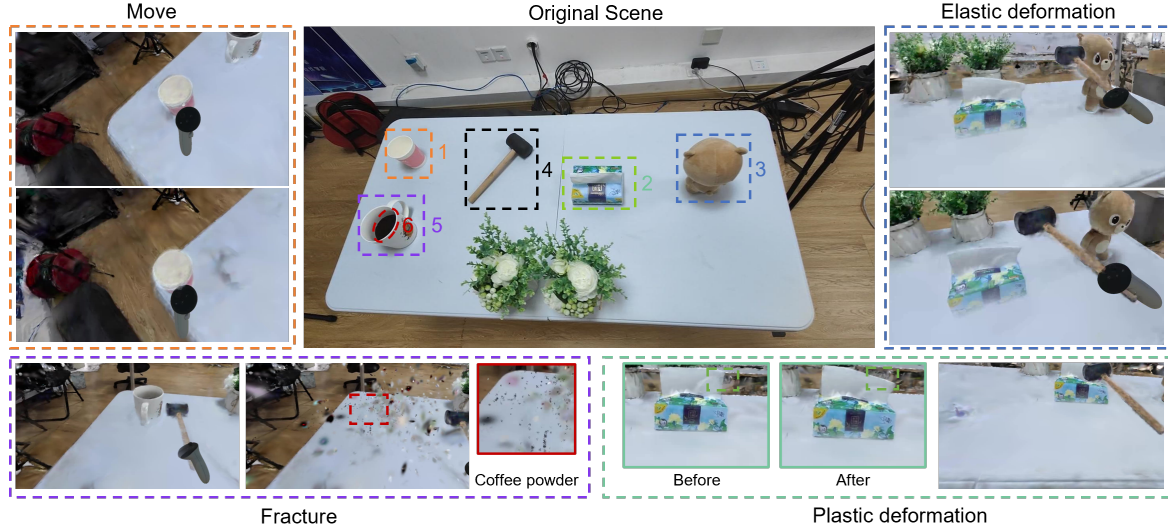


Figure 1: **Interaction in our custom dataset *labdesk*.** From one to six are respectively a paper cup, a box of tissues, a teddy bear, a plastic hammer, a mug, and coffee powder inside the mug. After segmentation and analysis, we use the hammer to interact with the other objects.

3 RESULTS

We evaluate our system through several classic datasets such as *wolf*, *sofa*, *fox*, and *garden*. We also include our custom data *labdesk* in real scenes captured with a mobile phone.

We compare our method with PhysGaussian to evaluate our real-time efficiency. We keep the simulation time and the iteration steps the same. As shown in Table 1, our method is faster.

Dataset	Frame time	Step time	PhysGaussian	Ours
wolf	40	0.02	4060	221
sofa	20	0.1	397	150
fox	40	0.1	837	273

Table 1: Performance (time to generate one frame (ms)) : ours and PhysGaussian.

In the meantime, it is unnecessary to use such a tiny step for PBD. We measure the number of Gaussian kernels and FPS in each demo as shown in Table 2, which demonstrates high efficiency.

Scene	Gaussian kernels	particles	FPS w/o simulation	FPS in simulation
wolf	188201	350	129.0	105.8
sofa	636507	1477	104.7	62.0
garden	1843725	262	64.3	58.1
labdesk	1024703	6826	68.1	36.4

Table 2: **Performance (FPS) in each scene.** We measure the number of Gaussian kernels and physical particles

For our material analysis based on LLM, we primarily show the results in *labdesk* due to the variety of objects in the scene. The analysis result is shown in Table 3 and the interaction results are shown in Figure 1.

4 CONCLUSION

We present LIVE-GS, a highly realistic interactive GS-based VR system with advanced scene understanding capabilities powered by LLM. From a few input images, our system enhances the scene by

Target	Category	Mass (kg)	Deformation Resistance	Plasticity
1	deformation	0.05	0.2	0.8
2	deformation	0.2	0.3	0.5
3	deformation	0.3	0.4	0.2
Target	Category	Mass (kg)	Force threshold (N)	Fragile
4	rigid	0.5	—	False
5	rigid	0.3	20	True
Target	Category		Friction	
6	granule		0.1	

Table 3: **Material analysis in our own dataset *labdesk*.** This scene contains various phenomena with different physical properties.

analyzing the physical properties of each element and tracking possible artifacts when reconstructing scenes. Even faced with complex real-world data, LIVE-GS supports various physical simulations for real-time interaction consistent with visual effects. Our work reveals the potential of the combination of 3DGS and LLM in dynamic scene reconstructions and interactions, which could become a mainstream method for future VR asset creation.

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